MA2040: Probability, Statistics & Stochastic Processes
Assignments 3&4

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Q 1. Let X be a random variable with probability density function

$$\rho_X(x) = \begin{cases} c(4-x^2), & \text{if } x \in [-2,2], \\ 0, & \text{if } x \notin [-2,2]. \end{cases}$$

(a) What is the value of the constant c?

(b) What is the cumulative distribution function $F_X(x)$ of X?

Q 2. The probability density function of the random variable X is given by

$$\rho_X(x) = \begin{cases} a+bx^2, & \text{if } x \in (0,1), \\ 0, & \text{if } x \notin (0,1). \end{cases}$$

Determine the values of the constants a and b if $E[X] = 35$.

Q 3. Do the following functions $F(x)$ represent cumulative distribution of some random variable? If yes, determine the corresponding probability density function.

(a)

$$F(x) = \begin{cases} 0, & \text{if } x \in (-\infty, 1), \\ j(j+1)/72, & \text{if } x \in [j, j+1), \text{ for all } j \in \{1,2,3,4,5,6,7\}, \\ 1, & \text{if } x \in [8, \infty). \end{cases}$$

(b)

$$F(x) = \begin{cases} 0, & \text{if } x \in (-\infty, -1), \\ 1/2, & \text{if } x \in [-1, 2), \\ 1, & \text{if } x \in [2, 4), \\ 3/2, & \text{if } x \in [4, \infty). \end{cases}$$

(c)

$$F(x) = \begin{cases} 0, & \text{if } x \in (-\infty, 0), \\ 1-e^{-x}+e^{-2x}, & \text{if } x \in [0, \infty). \end{cases}$$

(d)

$$F(x) = \begin{cases} 0, & \text{if } x \in (-\infty, 0), \\ 1 - 3e^{-x} + e^{-2x}, & \text{if } x \in [0, \infty). \end{cases}$$

Q 4. Determination of parameters

(a) Determine the constants a and b given that the function

$$F(x) = \begin{cases} 0, & \text{if } x \in (-\infty, -2), \\ a(x+2), & \text{if } x \in [-2, 2), \\ bx^2, & \text{if } x \in [2, 4), \\ 1, & \text{if } x \in [4, \infty). \end{cases}$$

is the cumulative distribution function of a random variable X with the additional information that $F(x)$ is continuous every where except at $x = 2$. The discontinuity of $F(x)$ at $x = 2$ is $1/8$.

(b) What is the minimum possible value of the parameter a such that

$$F(x) = \begin{cases} 0, & \text{if } x \in (-\infty, a), \\ 1 - 3e^{-x} + e^{-2x}, & \text{if } x \in [a, \infty). \end{cases}$$

is a cumulative distribution of some random variable?

Q 5. Let X be a random variable uniformly distributed in the interval $(0, 1)$. Calculate the probability density functions of the following functions Z of the random variables Xgiven below.

(a)

$$Z = \tan\left(\frac{\pi}{2}\left(X - \frac{1}{2}\right)\right), \text{ for } X \in (0, 1).$$

(b)

$$Z = \tan\left(\pi\left(X - \frac{1}{2}\right)\right), \text{ for } X \in (0, 1).$$

(c)

$$Z = \begin{cases} -1, & \text{if } X \in \left(0, \frac{1}{4}\right), \\ \tan\left(\pi\left(X - \frac{1}{2}\right)\right), & \text{if } X \in \left[\frac{1}{4}, \frac{3}{4}\right), \\ 1, & \text{if } X \in \left[\frac{3}{4}, 1\right). \end{cases}$$

(d)

$$Z = -a\log(X), \ a \text{ is a positive constant.}$$

Q 6. A system can function satisfactorily for a random amount of time T. If the probability density of T is given (in units of months) by

$$\rho_T(t) = \begin{cases} ct^2 e^{-t/2}, & \text{if } t \in (0, \infty), \\ 0, & \text{if } t \leq 0, \end{cases}$$

what is the probability that the system functions satisfactorily for at least 5 months?

Q 7. The probability density function of T, the lifespan of a certain type of electronic device (measured in hours), is given by

$$\rho_T(t) = \begin{cases} 20/t^2, & \text{if } t > 20, \\ 0, & \text{if } t \leq 20. \end{cases}$$

(a) Find Prob$(T > 40)$.

(b) What is the cumulative distribution function of T?

(c) What is the probability that, of 6 such types of devices, at least 3 will function for at least 30 hours? What assumptions are you making?

Q 8. A filling station is supplied with petrol once a week. If its weekly volume of sales (in units of ten thousand liters) is a random variable with probability density function

$$\rho_X(x) = \begin{cases} 10(1-x)^9, & \text{if } x \in (0, 1), \\ 0, & \text{if } x \notin (0, 1), \end{cases}$$

what must the capacity of the tank be so that the probability of the supply being exhausted in a given week is .01?

Q 9. Compute $E[X]$ and $var(X)$ if X is a random variable with probability density function given by

(a)

$$\rho_X(x) = \begin{cases} \frac{1}{4} x e^{-x/2}, & \text{if } x \in (0, \infty), \\ 0, & \text{if } x \leq 0. \end{cases}$$

(b)

$$\rho_X(x) = \begin{cases} c(1-x^2), & \text{if } x \in (-1, 1), \\ 0, & \text{if } x \notin (-1, 1). \end{cases}$$

(c)

$$\rho_X(x) = \begin{cases} 10/x^2, & \text{if } x \in (10, \infty), \\ 0, & \text{if } x \leq 10. \end{cases}$$

Q 10. Assume that the seasonal demand of a certain item is a continuous random variable X with its probability density ρ. If b is the net profit per unit that is sold and l is the net loss per unsold unit, what is the optimal stock of the item? Express the result in terms of the cumulative distribution function F by considering the expectation value of the profit.

Q 11. A point P is chosen at random on a line segment AB of length L so that AB is partitioned into two segments AP and PB. Find the probability that the ratio of the shorter to the longer of these segments (AP and PB) is less than 1/5.

Q 12. If X is a random variable with exponential probability density

$$\rho_X(x) = \begin{cases} e^{-x}, & \text{if } x \in (0, \infty), \\ 0, & \text{if } x \leq 0, \end{cases}$$

compute the probability density function of the random variable Y defined by $Y = log(X)$.

Q 13. If X is uniformly distributed over $(0, 1)$, find the probability density function of the random variable $Y = e^X$.

Q 14. Find the probability density function of the random variable $R = Asin(\theta)$, where A is a fixed constant and θ is uniformly distributed in the interval $(-\pi/2, \pi/2)$.

Q 15. If Y is uniformly distributed over $(0, 5)$, what is the probability that the roots of the equation $4x^2 + 4xY + Y + 2 = 0$ are both real?

Q 16. If X is uniformly distributed over $(1, 1)$, find

(a) $Prob(|X| > 1/2)$,

(b) the probability density function of the random variable $|X|$.

Q 17. Let T be the lifetime of a certain item with probability density function and the cumulative distribution function at $T = t$ are denoted by $\rho(t)$ and $F(t)$, respectively. Consider $Prob(T \in (t, t + dt) | T > t)$ which is the conditional probability that an item which is functional till t will not survive for an additional time dt. Prove that

$$Prob(T \in (t, t + dt) | T > t) = \frac{\rho(t)}{1 - F(t)} dt = \lambda(t) dt.$$

$\lambda(t)$ is called the hazard function.

Q 18. Suppose that the life distribution of an item has the hazard rate function $\lambda(t) = t^3$, for $t > 0$. What is the probability that

(a) the items lifetime is between 0.4 and 1.4?

(b] a 1-year-old item will survive to age 2?.

Q 19. In 10000 independent tosses of a coin, the coin landed on heads 5800 times. Is it reasonable to assume that the coin is not fair? Explain.

Q 20. A random experiment consists of choosing a point randomly from an annular disc with inner radius a and outer radius b (obviously, $a < b$). if R is the distance of this point from the center of the disc, what is the probability density function of the random variable R? What is the corresponding cumulative distribution function?

Q 21. Serious accidents involving two wheelers occur in a certain town at the rate of 20 per month. Take the number of days in every month to be 30.

(a) What is the probability that exactly 10 accidents will occur in the first 15 days of the month?

(b) Given that exactly 10 accidents occurred in the first 15 days what is the probability that they occurred in the last 5 days of these 15 days?

Q 22. Let T be the random variable corresponding to lifespan of a device with probability density given by

$$\rho_T(t) = \begin{cases} 0, & \text{if } t \in (-\infty, 0)\,, \\ \frac{1}{\tau} e^{-t/\tau}, & \text{if } t \in [0, \infty)\,. \end{cases} ,$$

where τ is a real positive constant. Show that $Prob\,(k \leq T < k+1)$ can be expressed as $(1-c)\,c^k$ with a suitable value for the constant c.

Q 23. Let X be a *normal* random variable with mean μ and standard deviation σ. Using the formula

$$\int_{-\infty}^{\infty} \exp\left(-ax^2 - bx\right) dx = \sqrt{\frac{\pi}{a}} \exp\left(\frac{b^2}{4a}\right),$$

if $Re\,(a) > 0$, prove that the characteristic function of the normal distribution is given by

$$g(k) = \int_{-\infty}^{\infty} e^{ikx} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) dx = \exp\left(ik\mu - \frac{k^2}{2}\sigma^2\right)$$

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Q 24. Let X be a random variable with probability density

$$\rho_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), x \in (-\infty, \infty)$$

Prove that probability density of the random variable $Z = (x-\mu)/\sigma$ is given by

$$\rho_Z(z) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{z^2}{2}\right)$$

. This will henceforth be called as the standard normal distribution.

Q 25. Let Z be a random variable with normal distribution $N\left(0, \sigma^2\right)$. Prove that

(a)
$$\left\langle Z^{2n+1} \right\rangle = 0, \text{ for all } n \in \mathbf{N}$$

(b)
$$\left\langle Z^{2n} \right\rangle = (2n-1)!!\sigma^{2n}, \text{ for all } n \in \mathbf{N}_{+},$$

where $(2n-1)!! = 1.3......(2n-1)$.

(c)
$$\langle z f(z) \rangle = \left\langle z^2 \right\rangle \left\langle \frac{df(z)}{dz} \right\rangle, \text{ if f (z) is a differentiable function of z}$$